

Ari Gestetner

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Education

Monash University | Melbourne, Victoria, Australia

Bachelor Science in Mathematics and Computational Science

2023-2025

- Awarded with summer and winter vacation research scholarships
- First place in an IMC algorithmic trading competition for Monash students hosted by Monash Association of Coding (MAC)

Experience

Teaching Associate (Intro to Programming With Python & Intro to Computer Science)

Feb 2024 - Present

Monash University

- Achieved 89% student satisfaction rating exceeding the faculty average of 82% by providing engaging, individualised and interactive classes and explanations
- Collaborated with fellow TAs to deliver applied classes, sharing feedback to drive continuous teaching improvements and enhance student learning outcomes

Research Scholarship – Dr. Buser Say

Jul 2023 - Feb 2025

Monash University

Python, SCIP, Cplex, Gurobi, Seaborn, MILP, Constraint Programming

- Benchmarked and presented different solvers and formulations finding results of over 2x decrease in solve time
- Investigated various linear programming constraint formulations based on neural network transition functions
- Refactored and extended [SCIPPlan](#) an automated planner for continuous time sequential planning problems over non-linear domains
- Assessed the performance of SCIPPlan in identifying optimal plans by formulating diverse problems with non-linear constraints
- Successfully derived feasible optimal solutions for a novel problem within a reasonable time frame
- Developed SCIPPlan+, an automated planner for discrete and continuous time planning problems with stochastic domains
- Researched the applications and performance of SCIPPlan+ over a variety of domains
- Introduced Github workflow to provide automated packaging and deployment to PyPi enabling a fast and easy development process
- Continued extending SCIPPlan to allow users to provide a system of odes to be solved and used in the planning process supporting engineers and physicists in their work
- Incorporated function interpolation, introducing flexibility to the planning process and allowing for more complex problems to be solved

Projects

Personal Website & Open Source Contribution

TypeScript, Nuxt, Vue, Vercel

- Developed a personal portfolio site using Nuxt, Vue and Obsidian for content editing, managing the full deployment lifecycle via Vercel and Github actions
- Contributed to the [Nuxt MDC](#) open source ecosystem to fix a bug relating to MathJax support, providing a fix to issues spread across multiple repositories

Finance API

Python, FastAPI, PyTest, BeautifulSoup

- Architected a high-performance API for stock data retrieval using FastAPI and serverless hosting
- Implemented multithreading for concurrent data scraping, resulting in a 5x increase in retrieval speed
- Integrated an automated testing suite using PyTest and engineered a Google Sheets add-on to manage \$15,000 in investments

Stock Paper Trader

Python, Flask, PostgreSQL, HTML, CSS

- Developed a full-stack CRUD application to facilitate imitation stock trading for investment education
- Implemented essential security best practices, including password hashing, to protect private user data

Home Server & Infrastructure

Docker, Reverse Proxies, Linux, Networking

- Manage a dedicated home server for self-hosting applications, gaining expertise in Docker deployment and container orchestration
- Configured vpn, reverse proxies and environments to enable secure and easy access and maintain high availability for hosted containers

Technical Skills

Languages: Python, R, Typescript, C, C++17, SQL, HTML/CSS

Frameworks: Flask, FastAPI, Django, PyTest, React, Vue, Nuxt, JUCE

Developer Tools: Git, Docker, RStudio, Linux, Vercel, GCP

Libraries: BeautifulSoup, Pandas, Matplotlib, Numpy/Scipy, Pytorch, Tidyverse, Fable, Cplex, Gurobi, SCIP

Academic Concepts: Analysis, Linear Algebra, Probability, Optimisation and OR, Graph Theory, Forecasting, Statistical Modelling, Econometric Theory